A Generalized Framework for Penalized Local Mode-Seeking: Unifying Mean Shift and Sparse Smoothing Splines

Swagato Das, Swagatam Das

April 25, 2026

Abstract

Mode-seeking algorithms, such as the classic Mean Shift, are foundational tools for non-parametric clustering and density analysis. However, standard formulations rely on zero-degree local constant estimators, which suffer from boundary bias and oversmoothing of high-curvature density peaks. In this report, we propose a generalized umbrella framework for mode-seeking, formulated as a Penalized Local Functional Maximization problem. We demonstrate that existing mode-seeking algorithms are special cases of this framework. Furthermore, we introduce a novel instantiation of this framework: the Sparse Smoothing Spline Mean Shift (SSS-MS). By employing a dual-penalty structure—an L_2 integral roughness penalty and an L_1 Group Lasso complexity penalty—the SSS-MS adaptively selects its own functional complexity at every spatial location, ensuring convergence properties that theoretically strictly dominate or equal the performance of its standard generalizations.

1 Introduction

Non-parametric mode-seeking algorithms aim to locate the local maxima of an underlying probability density function $f(x)$ from a set of empirical observations. The most ubiquitous of these, the Mean Shift algorithm, operates by performing gradient ascent on a Kernel Density Estimate (KDE).

While robust, standard Mean Shift is functionally rigid. It implicitly relies on a Nadaraya-Watson estimator, analogous to fitting a local constant to the density. Generalizing this to higher-order local polynomials or splines reduces bias but introduces severe risks of overfitting to local noise (Runge's phenomenon[1]) and requires rigid a priori selection of the polynomial degree.

Rather than proposing a narrow algorithmic fix to achieve state-of-the-art performance on specific benchmarks, this report constructs a generalized theoretical framework. By defining mode-seeking as an optimization over a regularized functional space, we encompass standard methods while introducing the mathematical machinery for adaptive complexity, culminating in the SSS-MS algorithm.

2 The Generalized Mode-Seeking Framework

Let $\mathcal{S} = \{x_i\}_{i=1}^n$ be a set of data points where $x_i \in \mathbb{R}^d$. We define the generalized mode-seeking trajectory of a point $x^{(t)}$ as the continuous tracking of the local maxima of an estimated spatial function $s(z; x^{(t)})$.

We formulate the local estimation at spatial location x as the minimization of a generalized empirical risk function over a hypothesis space $\mathcal{H}$:

[1] Runge's phenomenon is a problem in numerical analysis where high-degree polynomial interpolation using equally spaced points produces wild oscillations near the edges of an interval, causing the approximation to worsen rather than improve as more points are added.

$$\hat{s}_x = \arg\min_{s \in \mathcal{H}} [\mathcal{L}(\mathcal{S}, s, \mathbf{W}_x) + \mathcal{R}(s)] \tag{1}$$

Where:

- $\mathcal{L}$ is a local loss function (e.g., weighted least squares).
- $\mathbf{W}_x$ is a spatial weighting matrix defined by a kernel $K_h(x - x_i)$ with bandwidth h, governing the locality of the estimate.
- $\mathcal{R}(s)$ is a regularization functional that penalizes specific structural properties of s.

Once $\hat{s}_x$ is estimated, the spatial update (the "shift") is defined by a local optimization step, typically gradient ascent:

$$x^{(t+1)} = x^{(t)} + \eta \nabla \hat{s}_{x^{(t)}}(x^{(t)}) \tag{2}$$

3 Subsuming Existing Algorithms

This umbrella framework elegantly recovers standard mode-seeking algorithms through specific constraints on $\mathcal{H}$ and $\mathcal{R}$.

3.1 Standard Mean Shift

If we restrict the hypothesis space $\mathcal{H}$ to the space of constant functions ($s(z) = \beta_0$) and remove all regularization ($\mathcal{R}(s) = 0$), the local loss reduces to:

$$\min_{\beta_0} \sum_{i=1}^{n} (x_i - \beta_0)^2 K_h(x - x_i) \tag{3}$$

The solution is the weighted mean, and the shift vector reduces exactly to the classic Mean Shift vector.

3.2 Local Linear Mode-Seeking

If $\mathcal{H}$ is expanded to the space of linear functions ($s(z) = \beta_0 + \beta_1^T(z - x)$) with $\mathcal{R}(s) = 0$, the gradient update relies purely on $\hat{\beta}_1$, representing a localized first-order shift that reduces boundary bias but remains sensitive to noise.

4 The Sparse Smoothing Spline Mean Shift (SSS-MS)

We now instantiate the generalized framework to create an algorithm capable of adaptive complexity. We define $\mathcal{H}$ as the span of a hierarchical B-spline basis $B(z) = [B_1(z), \ldots, B_m(z)]^T$, partitioned into G groups representing increasing polynomial degrees.

To prevent overfitting and enable automatic degree selection without refitting, we define the regularizer $\mathcal{R}(s)$ as a dual-penalty mechanism:

$$\mathcal{R}(s) = \lambda_1 \int \|s^{(q)}(z)\|^2 dz + \lambda_2 \sum_{g=1}^{G} w_g \|\alpha_g\|_2 \tag{4}$$

Here, λ_1 controls the L_2 integral roughness penalty (enforcing analytic smoothness), and λ_2 governs an L_1 Group Lasso penalty (forcing the coefficients α_g of unnecessary higher-degree basis groups to exactly zero).

4.1 Optimization via Proximal Gradient Descent

Because the Group Lasso penalty is non-differentiable, the local coefficients $\hat{\alpha}$ cannot be found via closed-form normal equations. Instead, at each spatial step $x^{(t)}$, we employ the Iterative Shrinkage-Thresholding Algorithm (ISTA). The smooth unpenalized gradient step is augmented by the analytic penalty matrix $\mathbf{S}_{jk} = \int B_j^{(q)} B_k^{(q)} dz$, followed by a block soft-thresholding operator to enforce group sparsity.

Algorithm 1 SSS-MS Inner Coefficient Update

Require: Local weights $\mathbf{W}_x$, Basis $\mathbf{B}$, Penalty Matrix $\mathbf{S}$, λ_1, λ_2

Initialize $\alpha^{(0)}$
while not converged **do**
  $\nabla g \leftarrow -\mathbf{B}^T \mathbf{W}_x (\mathbf{X} - \mathbf{B}\alpha^{(k)}) + 2\lambda_1 \mathbf{S}\alpha^{(k)}$
  for $g = 1, \ldots, G$ **do**
    $\mathbf{z}_g \leftarrow \alpha_g^{(k)} - \gamma [\nabla g]_g$
    $\alpha_g^{(k+1)} \leftarrow \left(1 - \frac{\lambda_2 \gamma \sqrt{d_g}}{\|\mathbf{z}_g\|_2}\right)_+ \mathbf{z}_g$
  end for
  $k \leftarrow k + 1$
end while
return $\hat{\alpha} = \alpha^{(k)}$

5 Theoretical Guarantees

The primary advantage of this framework is its bounded performance guarantee relative to its special cases.

Theorem 1 (Bounded Empirical Risk): Let $\mathcal{E}_{ms}^*$ be the optimal empirical risk achieved by standard Mean Shift, and $\mathcal{E}_{sss}^*$ be the optimal risk of the SSS-MS. Because the constant function space is a strict subspace of the B-spline basis, and the regularization hyperparameters $\{\lambda_1, \lambda_2\}$ span $\mathbb{R}_{\geq 0}$, there exists a configuration such that $\mathcal{E}_{sss}^* \leq \mathcal{E}_{ms}^*$.

Furthermore, the inclusion of the strictly positive semi-definite matrix $\mathbf{S}$ guarantees that the local Hessian remains strictly diagonally dominant, ensuring stable, strictly convex optimization even in highly sparse spatial regions where standard local polynomial estimators become rank-deficient.

6 Conclusion

By casting mode-seeking as a generalized functional optimization problem, we provide a mathematically rigorous umbrella that unifies existing methods. The SSS-MS demonstrates how this framework can be leveraged to introduce adaptive spatial complexity, yielding an algorithm that strictly dominates or equals the theoretical performance of its predecessors while maintaining robust convergence guarantees.

A Appendix A: Proof of Theorem 1

Theorem 1. *Let $\mathcal{E}^*_{ms}$ be the optimal empirical risk achieved by standard Mean Shift, and $\mathcal{E}^*_{sss}$ be the optimal empirical risk of the Sparse Smoothing Spline Mean Shift (SSS-MS). There exists a hyperparameter configuration $\{\lambda_1, \lambda_2\}$ such that $\mathcal{E}^*_{sss} \leq \mathcal{E}^*_{ms}$. Furthermore, if the number of points with strictly positive kernel weights is at least q (where q is the order of the derivative penalized by $\mathbf{S}$), the SSS-MS local objective is strictly convex even if the local weight matrix $\mathbf{W}_x$ is rank-deficient.*

A.1 Bounded Empirical Risk (Subsumption)

Proof. Let $\mathcal{H}_{ms}$ denote the hypothesis space of standard Mean Shift, which is restricted to constant functions: $s(z) = \beta_0$. Let $\mathcal{H}_{sss}$ denote the hypothesis space of the SSS-MS, which is the span of a B-spline basis of degree p: $s(z) = \sum_{k=1}^{m} \alpha_k B_k(z)$.

By the partition of unity property of B-splines, $\sum_{k=1}^{m} B_k(z) = 1$ for all z within the domain. Therefore, any constant function $f(z) = c$ can be perfectly represented in $\mathcal{H}_{sss}$ by setting $\alpha_k = c$ for all $k \in \{1, \ldots, m\}$. Consequently, $\mathcal{H}_{ms} \subset \mathcal{H}_{sss}$.

The SSS-MS objective function is:

$$\mathcal{J}(\alpha; x) = \frac{1}{2}(\mathbf{X} - \mathbf{B}\alpha)^T \mathbf{W}_x (\mathbf{X} - \mathbf{B}\alpha) + \lambda_1 \alpha^T \mathbf{S} \alpha + \lambda_2 \sum_{g=1}^{G} w_g \|\alpha_g\|_2$$

Consider the asymptotic behavior as the regularization parameters $\lambda_1, \lambda_2 \to \infty$. The L_2 penalty matrix $\mathbf{S}$ penalizes the q-th derivative. As $\lambda_1 \to \infty$, any solution $\hat{\alpha}$ must lie in the null space of $\mathbf{S}$, which consists exactly of polynomials of degree strictly less than q. Simultaneously, the Group Lasso penalty $\lambda_2 \to \infty$ forces all coefficient groups corresponding to degrees ≥ 1 to identically zero.

Thus, in the limit, the optimal solution $\hat{\alpha}$ is forced into the space of zero-degree polynomials (constant functions), yielding $\mathbf{B}\hat{\alpha} = c\mathbf{1}$. In this limiting case, the SSS-MS objective collapses exactly to the standard Mean Shift objective:

$$\lim_{\lambda_1, \lambda_2 \to \infty} \min_{\alpha} \mathcal{J}(\alpha; x) = \min_{c} \frac{1}{2} \sum_{i=1}^{n} (x_i - c)^2 K_h(x - x_i) = \mathcal{E}^*_{ms}$$

Because the SSS-MS minimizes over a superset of the parameter space and $\{\lambda_1, \lambda_2\}$ span $\mathbb{R}_{\geq 0}$, the optimal regularized risk can never exceed the risk of the restricted (constant) model. Therefore, $\mathcal{E}^*_{sss} \leq \mathcal{E}^*_{ms}$. □

A.2 Strict Convexity under Rank Deficiency

Proof. The SSS-MS objective is strictly convex if and only if its smooth, unpenalized component $g(\alpha)$ is strictly convex, as the addition of the convex L_1 penalty does not violate convexity.

The smooth component is:

$$g(\alpha) = \frac{1}{2}(\mathbf{X} - \mathbf{B}\alpha)^T \mathbf{W}_x (\mathbf{X} - \mathbf{B}\alpha) + \lambda_1 \alpha^T \mathbf{S} \alpha$$

The Hessian of $g(\alpha)$ with respect to α is:

$$\mathbf{H} = \mathbf{B}^T \mathbf{W}_x \mathbf{B} + 2\lambda_1 \mathbf{S}$$

Because $\mathbf{W}_x$ is a diagonal matrix of non-negative kernel weights, and $\mathbf{S}$ is derived from an integral of squared derivatives, both matrices are symmetric and positive semi-definite (PSD). Therefore, $\mathbf{H}$ is at least PSD.

For $g(\alpha)$ to be strictly convex, $\mathbf{H}$ must be strictly positive definite (PD). This requires that for any non-zero vector $\mathbf{v} \in \mathbb{R}^m$, the quadratic form $\mathbf{v}^T\mathbf{H}\mathbf{v} > 0$. We will prove this by contradiction.

Assume there exists a non-zero vector $\mathbf{v}$ such that $\mathbf{v}^T\mathbf{H}\mathbf{v} = 0$. Because both components are PSD, this implies:

$$\mathbf{v}^T(\mathbf{B}^T\mathbf{W}_x\mathbf{B})\mathbf{v} = 0 \quad \text{and} \quad \mathbf{v}^T(2\lambda_1\mathbf{S})\mathbf{v} = 0$$

1. **From the integral penalty:** $\mathbf{v}^T\mathbf{S}\mathbf{v} = \int \left([\mathbf{B}(z)^T\mathbf{v}]^{(q)}\right)^2 dz = 0$. This implies that the q-th derivative of the function defined by $\mathbf{v}$ is zero everywhere. Therefore, the vector $\mathbf{v}$ must define a polynomial of degree strictly less than q. Let $P(z) = \mathbf{B}(z)^T\mathbf{v}$ be this polynomial.

2. **From the spatial weighting:** $\mathbf{v}^T\mathbf{B}^T\mathbf{W}_x\mathbf{B}\mathbf{v} = (\mathbf{B}\mathbf{v})^T\mathbf{W}_x(\mathbf{B}\mathbf{v}) = 0$. Because $\mathbf{W}_x$ is diagonal, this expands to $\sum_{i=1}^n w_i(P(x_i))^2 = 0$, where $w_i = K_h(x - x_i)$. This equation implies that $P(x_i) = 0$ for all points x_i where the kernel weight $w_i > 0$.

By the fundamental theorem of algebra, a non-zero polynomial of degree $d < q$ can have at most d roots (and therefore at most $q - 1$ roots). If the number of local points with strictly positive weights $w_i > 0$ is at least q, then $P(x_i)$ evaluates to zero at q or more distinct locations.

The only polynomial of degree $< q$ that has $\geq q$ roots is the trivial zero polynomial, $P(z) = 0$ for all z. For a B-spline basis, $B(z)^T\mathbf{v} = 0$ implies $\mathbf{v} = \mathbf{0}$.

This contradicts the assumption that $\mathbf{v}$ is a non-zero vector. Therefore, the null spaces of $\mathbf{B}^T\mathbf{W}_x\mathbf{B}$ and $\mathbf{S}$ intersect only at the origin. Consequently, $\mathbf{H}$ is strictly positive definite, and the local estimation problem is strictly convex. □

B Appendix B: Finite Sample Generalization Bound

To establish the reliability of the SSS-MS algorithm in practical regimes, we derive a finite sample bound on the local empirical risk. Let the true, unknown underlying probability measure of the data be P. We define the expected (population) risk at spatial location x as:

$$R_x(s) = \mathbb{E}_{X\sim P}[\mathcal{L}(X, s, \mathbf{W}_x)]$$

where $\mathcal{L}$ is our local weighted least squares loss. Our SSS-MS algorithm minimizes the regularized empirical risk $\hat{R}_{x,n}(s) + \mathcal{R}(s)$ over a finite sample of size n.

Let $\mathcal{H}_\Lambda$ be the effective hypothesis space restricted by the dual regularization penalties, such that for any $s \in \mathcal{H}_\Lambda$, the roughness is bounded by $\|s^{(q)}\|_2^2 \leq C_1(\lambda_1)$ and the group complexity is bounded by $\sum w_g\|\alpha_g\|_2 \leq C_2(\lambda_2)$.

Theorem 2 (Finite Sample Excess Risk Bound). *Assume the spatial domain is bounded, the local weight kernel K_h is bounded by B_K, and the B-spline basis functions are bounded such that the loss function is L-Lipschitz continuous with respect to the function outputs, taking values in $[0, M]$. Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of an i.i.d. sample $\mathcal{S}$ of size n, the following holds for all $s \in \mathcal{H}_\Lambda$:*

$$R_x(s) \leq \hat{R}_{x,n}(s) + 2L \cdot \mathfrak{R}_n(\mathcal{H}_\Lambda) + M\sqrt{\frac{\ln(1/\delta)}{2n}}$$

where $\mathfrak{R}_n(\mathcal{H}_\Lambda)$ is the empirical Rademacher complexity of the restricted hypothesis class.

B.1 Proof of Theorem 2 and the Advantage of the L_1 Penalty

Proof. Because the loss function $\mathcal{L}$ is bounded in $[0, M]$ and changing a single data point x_i in the sample $\mathcal{S}$ can change the empirical risk by at most M/n, we apply McDiarmid's Inequality. Let $\Phi(\mathcal{S}) = \sup_{s \in \mathcal{H}_\Lambda}(R_x(s) - \hat{R}_{x,n}(s))$. With probability at least $1 - \delta$:

$$\Phi(\mathcal{S}) \leq \mathbb{E}_{\mathcal{S}}[\Phi(\mathcal{S})] + M\sqrt{\frac{\ln(1/\delta)}{2n}}$$

By a standard symmetrization argument introducing Rademacher variables $\sigma_i \in \{-1, +1\}$, the expectation of the supremum is bounded by twice the expected Rademacher complexity of the loss class. By the Ledoux-Talagrand contraction lemma for L-Lipschitz loss functions, this is bounded by $2L \cdot \mathfrak{R}_n(\mathcal{H}_\Lambda)$. □

Remark on Complexity Scaling: The power of the SSS-MS lies in the scaling of $\mathfrak{R}_n(\mathcal{H}_\Lambda)$. Because the Group Lasso penalty restricts the $L_{2,1}$ norm of the coefficients, the Rademacher complexity is heavily constrained by the active groups. If the basis matrix $\mathbf{B}$ has dimension m, a standard L_2-only penalty (like standard P-splines) yields a complexity scaling of $\mathcal{O}(\sqrt{m/n})$.

However, by enforcing group sparsity via λ_2, the complexity scales logarithmically with the ambient dimension of the basis:

$$\mathfrak{R}_n(\mathcal{H}_\Lambda) \leq \mathcal{O}\left(C_2(\lambda_2)\sqrt{\frac{\log m}{n}}\right)$$

Therefore, the SSS-MS allows the practitioner to define an arbitrarily complex, high-dimensional hierarchical basis structure $m \gg n$ without suffering a catastrophic degradation in finite sample generalization, provided the true underlying density manifold can be represented sparsely. This guarantees that the estimated spatial gradient vector $m(x^{(t)})$ tightly bounds the true population gradient, ensuring stable mode-seeking even in low-sample regimes.